GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	**Specification for Pipeline Construction**
Document Number	EA-BFPCL-GPMC-GGPL-FEED- PPL-SPC-009
Document Revision	A01
Document Status	Issued for Approval
Originator	E. Uwaoma
Security Classification	Restricted
Issue Date	13-Dec-22

Specification for Pipeline Construction

A01	13.12.22	Issued for Approval	E. Uwaoma	E. Nwachukwu	A. Duru	
R01	25.05.22	Issued for Review	E. Uwaoma	E. Nwachukwu	A. Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	25.05.22	First Issue for Clients Review
A01	13.12.22	Issued for Approval

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited		
Project:	Gas Processing & Methanol Complex	**Project No.** 11998A	
Facility:	Gas Gathering Pipelines & Associated Infrastructure	**Doc No.** EA-BFPCL-GPMC-GGPL-FEED- PPL-SPC-009	

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES	
Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Wall Thickness Calculation Report
EA-BFPCL-GPMC-GGPL-FEED-PPL-ALS-001 to 036	Pipeline Alignment Sheet
EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-010	Specification for Pipeline Cleaning, Gauging and Hydrotesting
EA-BFPCL-GPMC-GGPL-FEED-PPL-PHY-001	Pipeline Integrity Management Philosophy.
EA-BFPCL-GPMC-GGPL-FEED-PPL-PHY-002	HDD Installation Philosophy.

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Abbreviations

Abbreviation	Meaning
ASME	American Society of Mechanical Engineers
API	American Petroleum Institute
CS	Carbon Steel
DEP	Design and Engineering Practices
DP	Design Pressure
DWT	Drop Weight tear
DCVG	Direct Current Voltage Gradient
FEED	Front End Engineering
HAZ	Heat affected zone
ISO	International Organisation for Standardization
LTO	License To Operate
3LPE	Three-Layer Polyethylene
MMscf/d	Million standard cubic feet per day
NDE	Non-Destructive Examination
NPS	Nominal Pipe Size
OD	Outer Diameter
PE	Polyethylene
PSL	Product Specification Level
TD	Design Temperature
SMTS	Specified Minimum Tensile Strength
SMYS	Specified Minimum Yield Strength
WT	Wall thickness
WPS	Welding Procedure Specification
WPQR	Welding Procedure Qualification Record

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

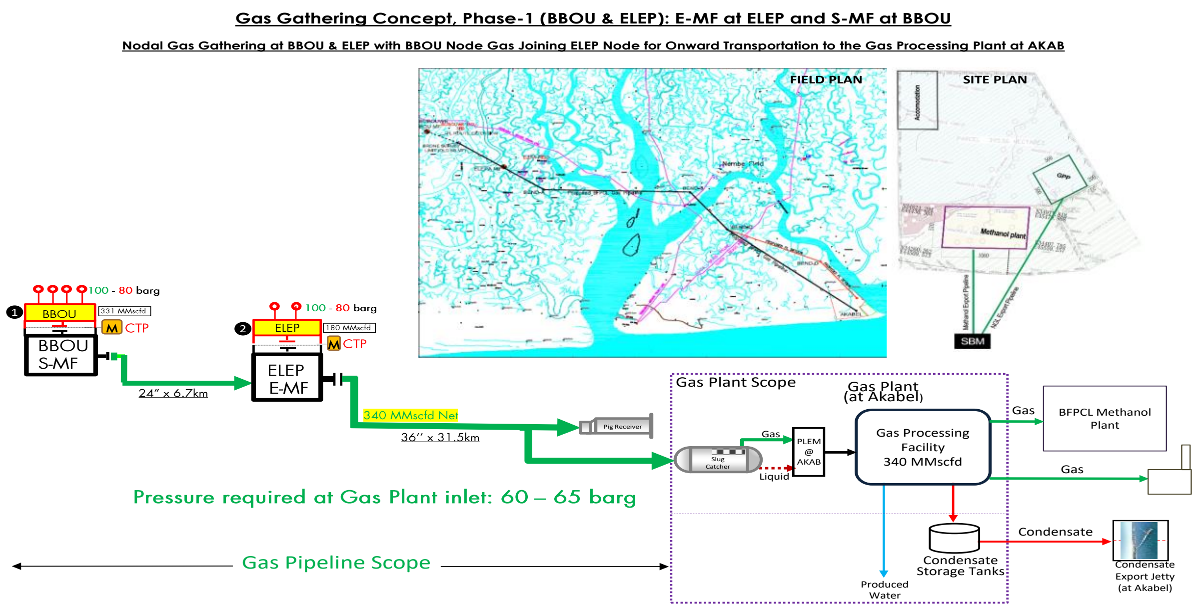

Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This document presents the specific and minimum requirements for the construction of the mainline of phase 1 of BFPCL Gas Gathering Pipelines & Associated Infrastructure. The line pipe are produced in accordance with the requirements of the latest edition of API Spec 5L PSL 2 for a multi-phase.

This construction specification provides the assessment issues related to the pipeline construction along the selected pipelines ROW and to recommend construction methods.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
DEP. 31.40.20.37-Gen	Line pipe for Critical Service (amendments/supplements to IOGP S-616 AND DNVGL-ST-F101 2019)
DEP. 82.00.10.10-Gen	Project Quality Assurance
DEP. 31.40.40.38-Gen	Hydrostatic Pressure Testing of new Pipeline (February 2017)
DEP. 61.40.20.30-Gen	Welding of Pipelines & Related Facilities (Amendments / supplements to API Std 1104)
ASME B31.8	Gas Transmission and Distribution Piping Systems
ASME B36.10M	Welded and Seamless Wrought Steel Pipe
API 5L	Specification for Line Pipe
ASME B36.10M	Welded and Seamless Wrought Steel Pipe
API RP 5LW	Recommended Practice for Transportation of Line pipe on Barges and Marine Vessels
API RP 5L1	Recommended Practice for Railroad Transportation of Line Pipe
API 1104	Standard for Welding Pipelines and Related Facilities
API RP 5L6	Transportation of Line Pipe on Inland Waterways
ASNT SNT-TC-1A	Personnel Qualification and Certification in Non-Destructive Testing
API 1102	Steel Pipeline Crossing Railroad and Highways
ISO 3183: 2007	Petroleum and Natural Gas Industries Steel Pipe for pipeline transportation systems.
ISO 9001	Quality Systems Model for Quality Assurance in Design, Development, Production, Installation and Servicing
ISO 13623	Petroleum and Natural Gas Industries - Pipeline Transportation

	Systems
ISO 3183	Petroleum and Natural Gas Industries – Steel Pipe for Pipeline Transportation Systems
ISO TS 29001	Petroleum, Petrochemical and Natural Gas Industries – Sector Specific Quality Management Systems Requirements for Product and Service Supply Organizations

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Line Pipe Data

3.1. 24” Multiphase Pipeline Data

The 24” pipeline design data for the Multiphase Pipeline Project are as shown below

Table 4.1: 24” Multiphase Pipeline Data

Parameter	Description
Design Code	ASME B31.8
From	BBOU Manifold Station
To	ELEPA Manifold Station
Pipeline Length	6.7km
Design Life	30 years
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature (Vent Line)	-13.3°C
Line Pipe Material Grade	API 5L–X70(PLS2) SAWL - 100% radiographed.
Installation Temperature	23°C
Specified Minimum Yield Strength (SMYS)	483 Mpa
NPS	**24"**
Density	7,850 kg/m3
Young’s Modulus of elasticity	2.07 x 105 MPa
Poisson’s ratio	0.3
Coefficient of linear thermal expansion	11.7 x 10-6 m/ m/ °C
Ultimate Tensile Strength	570 MPa
Steel Specific Resistivity	0.018 Ω m/mm2
OD / Wall Thickness	**15.88mm**
Design Factor	0.6
Corrosion Allowance	**3 mm**
ANSI Class	900#
Parent Pipe External Coating	3LPE
Concrete Coating Thickness	**60mm**

Installation Condition	Trenching
Cathodic Protection	Impressed current

3.2. 36" Multiphase Pipeline Design Data

The 36" pipeline design data for the Multiphase Pipeline Project are as shown below

Table 4.2: 36" Multiphase Pipeline Data

PARAMETER	DESCRIPTION
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Design Life	30 years
Process Fluid	Condensate/hydrated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature	-13.3°C
Line Pipe Material Grade	API 5L–X70 (PLS2) SAWL - 100% radiographed.
Installation Temperature	23°C
Specified Minimum Yield Strength	483 Mpa
NPS	**36"**
Density	7,850 kg/m3
Young's Modulus of elasticity	2.07 x 105 MPa
Poisson's ratio	0.3
Coefficient of linear thermal expansion	11.7 x 10-6 m/ m/ °C
Ultimate Tensile Strength	570MPa
Steel Specific Resistivity	0.018 Ω m/mm2
OD / Wall Thickness	**23.83mm**
Design Factor	0.6
Corrosion Allowance	**6 mm**
ANSI Class	900#
Parent Pipe External Coating	3LPE
Concrete Coating Thickness	**85mm**
Installation Condition	Trenching
Cathodic Protection	Impressed current

4. Design and Construction Conditions

4.1. General

The Mainline system comprises of two segments.

4.1.1. 24" x 6.7 Km Pipeline System

- The 24" x 6.7.Km pipeline system departing the BOUBUWE (BBOU) Manifold Station to ELEPA (ELEP) manifold station comprises of the following,
- 24" x 28" Pig launcher and its appurtenances located at BBOU manifold station.
- 24" Double Block and Bleed arrangement at BBOU,
- 24" The Safety shut down valves at BBOU
- The 24" Isolation Joint at BBOU
- Sacrificial Anode / Bracelet Anode between BBOU and ELEPA
- Fibre Optic Cable between BBOU and ELEPA
- Cathodic Protection System between BBOU and ELEPA
- The 24" Isolation Joint at ELEPA
- The 24" Safety shut down valves at ELEPA
- 24" Double Block and Bleed Valve arrangement at ELEPA,
- 24" X 28" Pig Receiver and its appurtenances located at ELEPA manifold station.

4.1.2. 36" x 31.5 Km Pipeline System

- The 36" X 31.5" Km pipeline system departing the ELEPA manifold station to AKEBEL (AKAB) station comprises of the following,
- 36" X 40" Pig launcher and its appurtenances located at ELEPA manifold station.
- 36" Double Block and Bleed arrangement at ELEPA,
- The 36" Safety shut down valves at ELEPA
- The 36" Isolation Joint at ELEPA
- Sacrificial Anode / Bracelet Anode between ELEPA and AKABEL
- Fibre Optic Cable between ELEPA and AKABEL
- Cathodic Protection System between ELEPA and AKABEL
- The 36" Isolation Joint at AKABEL
- The 36" Safety shut down valves at AKABEL
- 36" Double Block and Bleed Valve arrangement at AKABEL,
- 36" X 40" Pig Receiver and its appurtenances located at ELEPA manifold station.

4.2. Codes for Construction

The construction of the pipeline shall be undertaken in accordance to DEP 31.40.00.10-Gen-Pipeline Engineering, DEP 31.40.10.20-Gen Flexible Composite Pipe Systems and this specification. In addition, construction shall be in conformity with the documents, regulations, codes, and standards as referred to in section3 of this document.

5. Construction Procedures

5.1. General

Detailed calculations, procedures, drawings, etc. shall be prepared by CONTRACTOR and submitted to EMPLOYER for approval to ensure that the pipelines will be installed in a safe and timely manner and with minimum impact on the environment before construction work starts.

The construction shall be executed in compliance with relevant laws and regulations and in accordance with contractual and project requirements. Calculations, procedures, drawings shall be prepared according to project specifications and EMPLOYER's specifications.

The construction procedures, as a minimum must cover the activities listed hereunder. List of construction procedures / method statements to be supplied by CONTRACTOR:

- Site Office & Facilities
- Method of Liaison with Landowners & Authorities
- Safety Plan
- Progress Reporting
- Construction Equipment List
- Line pipe Unloading, Receiving, Transport, Handling and Storage
- Fittings, tees, bends unloading, receiving, transport, handling, storage
- Survey Method
- Clearing & Setting Out
- R.O.W. Preparation in Normal Terrain
- R.O.W. Preparation in Swamp Areas
- Pipe Handling, Stringing & Internal Inspection
- Trenching
- Welding Procedure specification and Qualifications
- Welders Qualification
- Consumables for welding: Receiving, Handling, Storage
- Pipe Cold Bending
- Pipe Welding (monitoring)
- Monitoring & Mitigation of AC Induced Voltages during Pipeline Construction
- Weld Numbering
- ARC Burn Repairs (cut out and replacement)
- Non-Destructive Testing
- Field Weld Joint Coating

- Field joint coating Consumables: Receiving, Handling, Storage
- Normal Terrain-Lowering In
- Swampy Area-Lowering In
- Road Crossings (if any)
- Pipeline Crossings (if any)
- Instrumentation & Controls
- Works at/or near Operating Facilities
- Tie-ins (if any)
- Cleaning, Gauging & Hydrostatic Testing
- Cathodic Protection
- Re-Instatement
- Pre-commissioning
- As-Built Survey
- As-Built Documents & Drawings

CONTRACTOR shall verify that he has copies of the latest editions of all relevant Design documents and drawings specially all which are implied or referenced in this specification. CONTRACTOR shall give due consideration to specific notes as stated in documents and stipulated on drawings. CONTRACTOR shall subsequently revise these Contract documents and drawings as applicable with as built construction information.

5.2. Permit to Work

CONTRACTOR shall develop a permit to work system and maintain a Permitting Matrix for any work to be carried out involving interfaces with any 3rd parties (e.g. all other CONTRACTORs associated with the project and existing facilities). This system shall be subject to the approval of the EMPLOYER. No work shall be carried out in areas where interfacing is an issue without a valid permit.

CONTRACTOR shall secure, without cost to EMPLOYER, any official permits, relevant to the Construction Work, including permits necessary for the transportation of pipe and other materials; crossing of roads watercourses and all other permits that may be required in connection with the Construction Works.

5.3. Planning Approval

The CONTRACTOR shall not enter upon any land for setting out, clearing, and establishing the right-of-way without the written permission of the EMPLOYER's

6. Health Safety and Environment

Health, Safety and Environmental constraints and requirements shall take precedence over all specifications.

CONTRACTOR shall show evidence of a suitable Health, Safety and Environment Management System, conforming to the applicable requirements and statutory legislation. The EMPLOYER shall retain the right to inspect all equipment, plant related facilities. CONTRACTOR shall carry out a Hazard Analysis to identify and assess hazards associated with every work/activity.

Prior to commencement of any work within a caged facility, all gates must be unlocked and opened to enable personnel to escape in case of emergency. On completion of each work period the gates shall be securely locked.

The environmental sensitivity of the pipelines requires extensive monitoring to ensure minimal environmental disruption. Precautions shall be taken by the CONTRACTOR to avoid spillages of any substance, waste, or refuse. A Pollution Contingency Plan shall be submitted to the EMPLOYER for approval prior to commencement of construction work activities.

All risks to personnel and the environment resulting from construction works shall be As Low As Reasonably Practicable (ALARP).

6.1. Site Emergency Plan

Prior to mobilization, the CONTRACTOR shall submit to the EMPLOYER a Site Emergency Plan. The basic requirement of the plan is to describe how the CONTRACTOR plans to respond appropriately to an emergency.

7. Equipment and Personnel

Full technical specifications, descriptions and information of all plant and equipment, which CONTRACTOR intends to deploy for the construction, including the locations of deployment, shall be submitted to the EMPLOYER for review and approval. Technical descriptions and information shall also indicate the quantity of each type of equipment that will be utilized, including redundancy. All plant and equipment shall be appropriate and fit for purpose and maintained in sound operational condition.

Relevant certification and test certificates for cranes, winches and other plant and equipment shall be maintained in a valid status throughout the construction period. Maintenance or repairs (with exception of emergencies) shall only be undertaken at designated locations where safety and environmental requirements are met. Maintenance or repairs at the construction site shall not be permitted. Only spare parts approved by the manufacturer of plant and equipment shall be utilized. The CONTRACTOR shall provide enough of all necessary spare parts to keep all equipment operational.

Key plant and equipment shall be fully tested prior to mobilization to prove that the requirements of the work will be met. The EMPLOYER reserves the right to witness tests of key equipment. Prior to receipt of EMPLOYER's approval equipment shall not be mobilized to site. Plant and equipment shall only be demobilized from the construction site after final completion of its tasks or it is being replaced by an equivalent or upgraded item of equipment. In all cases a certificate of approval from the EMPLOYER must be issued before the item of plant leaves the site.

All equipment used for handling of coated pipe, fittings and other materials shall be adequately padded to protect coating and bevels of the material. A spreader bar lifting tackle of a type approved by the EMPLOYER for lifting of coated pipe shall be used.

The CONTRACTOR shall develop and submit a project organization chart for approval by the EMPLOYER. The organization chart shall indicate key personnel from project manager down to construction foremen and lines of authorization and communication. Curriculum vitae of these key personnel shall be submitted to the EMPLOYER for acceptance and approval of the individual key personnel. All key personnel shall have a good command of the English language verbal and written. In addition, the CONTRACTOR shall prepare a list of all personnel the CONTRACTOR intends to employ for the execution of the construction. All personnel shall be skilled and trained for their function and position. The EMPLOYER reserves the right to reject personnel or plant and equipment if in its opinion the respective person or plant and equipment is inadequate or unsuitable to fulfil the intended task.

Throughout the construction period, the CONTRACTOR shall maintain detailed records of personnel, plant and equipment employed for the construction. The lists shall indicate as minimum:

- Name;
- Type;
- Identification number;
- Date of mobilization;
- Operational status;
- Planned date of demobilization;

7.1. Periods of absence.

Other records are to be included as deemed necessary. These lists / records shall be submitted on a weekly

8. Materials Supply, Handling and Storage

8.1. Material Documentation

All materials and equipment shall be subject to a proper documentation filing procedure prepared by CONTRACTOR and approved by the EMPLOYER.

8.2. Pipe Handling

The handling of pipe and all other materials (loading, hauling, stacking, storage, etc.) shall be performed in such a manner that the pipe and other materials are adequately always protected from damage or loss. Recommendations of material CONTRACTORs shall be strictly followed.

Special care shall be taken to ensure that no soil, debris, muddy water, dirt, extraneous material, etc. enters the pipe and other material. Pipe and other materials shall be sealed by means of respective covers and / or caps.

The CONTRACTOR shall provide and maintain all necessary slings and lifting equipment suitable for properly handling of pipes, to the approval of the EMPLOYER Representative.

The booms of all side booms must be padded with rubber or an approved material to avoid damage to pipe coating in case of contact between pipe and boom.

All lifting equipment and slings shall be subject to testing/certification and regular inspection before use. Adequate colour coding/marking of the equipment will have to be provided by CONTRACTOR in compliance with CONTRACTOR's Lifting Procedure.

Lifting slings shall be marked with the safe working load to avoid being used above the specified permissible load.

Pipes shall not be subjected to jars or impacts and shall be lifted or lowered from one level to another as described below:

For pipes 6-ins (150mm) or more in diameter, special end hooks, curved to fit the inside of the pipe ends shall be used for lifting. In the case of pipes having end caps, band slings shall be used. Punching of end caps shall not be permitted.

Pipes shall be lowered into position in single lengths without dropping, and each length shall nest evenly with the lengths previously stacked or loaded.

Coated and wrapped pipe shall be handled at all times with approved rubber-covered broad band slings or other equipment designed to prevent damage to the coating and wrapping. The use of tongs, grabs, crowbars, chain slings, wire-rope and rope slings (of any sort), narrow or defective belt slings, or any other handling equipment which may be injurious to the pipe coating and wrapping, shall not be permitted. All joints and sections of coated and wrapped pipe shall be picked up clear of the ground and not dragged over the ground. Coated pipe shall not be subjected to jarring impact and shall be set down on well-padded skids. Walking on coated pipe shall not be permitted.

8.3. Transport of Materials and Equipment

CONTRACTOR shall prepare a transportation management plan subject to approval of EMPLOYER. This plan shall consider transport infrastructure conditions from point of material origin to its destination at site. The plan shall include, but not be limited to, types of transport vehicles, loads to be transported, register of all roads to be used, current status of roads to be utilized, load limitations of roads and bridges, any roads that cannot be used by certain types of vehicles, legal load requirements allowed on access roads, bridges and culverts and any height restrictions applicable, authorization and security requirements.

CONTRACTOR shall load/unload and haul all materials, supplies, machinery, and equipment to be used for the construction of the pipeline, from delivery points to and from field stores and to points along the right-of-way where all such items are to be used, whether the items are furnished by the EMPLOYER or the CONTRACTOR.

Loading, hauling, and unloading shall be performed in such a manner as to prevent damage to pipe and other materials; and, if any pipe or materials do sustain damage, the CONTRACTOR shall be responsible for all repair work and/or replacement costs.

Line pipe and all other materials shall be transported by vehicles in a manner approved by the EMPLOYER Representative. When pipe is being transported, the bottom layer shall be carried in shaped cradles, lined with protective rubber or similar approved padding. Subsequent layers of pipe shall be suitably padded in nesting on the previously loaded layer of pipe. The height of the stack on the vehicle shall not exceed the limits specified by DEP 31.40.30.30-Gen. If the pipe is being transported mounted in pyramid fashion, bolsters shall be required under the bottom layer with an approved yoke over the top layer and with a binding system such as to avoid damage to the pipe. Any stanchions used by pipe transportation to cage pipe loads shall require CONTRACTOR to ensure calculations are provided to determine tensile strength, axial load and shear strength levels have a 50% safety factor applied.

On completion of loading, pipes and all other items shall be adequately secured in such a manner as to prevent damage to the pipe coating and wrapping during transportation.

8.4. Stacking and Storage

To prevent damage to the protective coating, coated and wrapped pipes shall not be placed on the ground, but set down on timbers not less than 230mm wide and 150mm thick, with approved soft padding and wedges. The padding shall overhang the width of the timbers by not less than 25mm each side and have a minimum thickness of 50mm before load compression. The timbers, padding and wedges shall be provided and placed by the CONTRACTOR.

Where stacking is necessary, pipes shall be stacked no higher than 3 stacks for the 24” and 36” line pipes. Maximum stacking height of concrete-coated pipes shall be 4 m (13 ft) unless local regulations are more stringent.

At stockpile sites for pipe storage, the pipe shall be placed on stone free soil berms covered in heavy gauge polythene or geo-textile material to prevent washout during inclement weather. Adequate drainage channels shall also be installed to allow water to flow to a suitable discharge point and in turn mitigate any possible berm erosion. Pipe stacks shall be segregated according to type and pipe diameter.

Any hay or straw or other combustible material which has been used for packaging purposes shall be promptly removed from the Site by the CONTRACTOR. No waste packaging materials shall be burnt on the Site.

8.5. Handling of Materials other than Line Pipes

Materials that are liable to deterioration or damage, due to humidity, exposure to extreme temperatures or other adverse weather condition, shall be stored indoors and suitably protected.

CONTRACTOR shall strictly comply with CONTRACTOR instructions regarding the maximum storage temperatures and conditions of all materials, especially materials that are liable to change in primary characteristics due to unsuitable storage.

Materials other than pipe materials such as, but not limited to, valves, fittings, scraper traps etc., shall be lifted by suitable equipment provided by CONTRACTOR at all locations. When lifting lugs are provided on materials, suitable lifting hooks shall be used for lifting the materials. It is prohibited to use delicate parts such as hand wheels, nozzles, etc for lifting of the materials.

8.6. Material Inventory

CONTRACTOR shall maintain adequate method of inventory and material transfer that will be used to provide accurate accountability of all materials during the Construction Works.

9. Construction and Installation

9.1. Survey and Positioning for Construction

The CONTRACTOR shall perform all necessary field surveys for the establishment of the pipeline trench centre line, any crossings, proper grading, and bends. CONTRACTOR shall determine the exact location of all existing underground facilities and shall take all necessary measures to ensure that they are protected from damage. CONTRACTOR shall make every effort to locate, mark and report any third-party pipelines or services.

9.2. Right of Way Preparation

9.2.1. Contractual Obligations

- The pipeline ROW shall exactly follow the route surveyed and provided by EMPLOYER to the CONTRACTOR. All deviations shall be specifically agreed in writing between EMPLOYER and CONTRACTOR.
- CONTRACTOR shall be responsible for ensuring that EMPLOYER's survey pillars and any marker/pillar of other Companies and Proprietors are maintained and are not damaged or destroyed by his personnel or equipment.
- The work shall be performed in such a way that all operations can be carried out within the R.O.W.

9.2.2. Access to the Right of Way

- CONTRACTOR shall not enter any land for construction purposes without the written permission of EMPLOYER.
- Access across the ROW. strip during construction shall be provided, where resources or facilities are rendered inaccessible.
- Provisions shall be made to fence the ROW as necessary to prevent animals falling into the pipe trench were deemed necessary. CONTRACTOR shall liaise with the Proprietors to identify the areas that may require temporary fencing. Agreement as to the type of temporary fencing shall be reached with the Proprietors.

9.2.3. Clearing of Row and Care of Existing Features

- Clearing of the ROW shall be performed in such a manner so that it does not affect the schedule of subsequent operations. Adequate plant, equipment and manpower shall be provided to ensure the fulfilment of the work schedule for the Construction Works.
- EMPLOYER and each Proprietor and/or Parties having rights over the properties connected with the ROW, shall be notified by CONTRACTOR prior to the commencement of the

clearing and grading works.

- CONTRACTOR shall remove any obstacles, which may hinder the installation of the pipeline and related facilities. To carry out such work, CONTRACTOR shall adopt all necessary precautions and shall use the most suitable equipment to prevent damage and to reduce as much as possible demolition works.
- Where the ROW crosses forest and tree plantations, any trees (irrespective of their type or size) which have to be cleared before grading operations, shall be cut at their base. They shall then be stripped and cut into lengths of 2.0m to 4.0m, or as specified by the Proprietors', their stumps shall be uprooted, and everything shall be stacked at the edges of the working area. When cutting down tall trees their fall shall be controlled in a safe manner on to the working strip.
- CONTRACTOR shall remove all brush and other loose debris so that the spoil material from the trenching operations shall not fall on any foreign material that might become mixed with the excavated soil. CONTRACTOR, under his Community Relations Management System, shall offer any removed timber destroyed during the construction activities to the local communities.
- Vegetation should be cleared in such a manner that it falls into the area within the ROW or approved working space boundaries and away from watercourses.
- Clearing and construction debris which may obstruct watercourses flow hinder fish passage, contribute to flood damages or result in watercourses bed scour or erosion shall be removed and properly disposed of.
- CONTRACTOR shall clear and dispose off standing crops, bush, vegetation, fences, (stock-proof temporary gates shall be installed in fences crossing the ROW. and kept closed at all times except when transiting through or with agreement from landowners, non-saleable timbers, tree roots and debris within the acquired boundaries of the ROW or on extra land acquired temporarily for construction use.
- When the ROW crosses woodlands or areas where easily combustible herbaceous material is present, all necessary precautions shall be taken to prevent uncontrolled fires.

9.2.4. Waste Material

- CONTRACTOR shall contain all forms of debris from his operation, such as cans, bottles, containers, welding rods, etc., and dispose of such debris at approved waste disposal sites. All such rubbish to be removed from ROW. at close of business at latest every day.
- Topsoil Stripping and Consideration
- In dry land locations, CONTRACTOR shall strip the layer of topsoil or organic surface material across the full working width. Topsoil shall be stripped to a depth of 300 mm maximum

unless otherwise agreed or specified on construction drawings.

- The topsoil and the debris shall be stored at the designated dumpsites along the edge of the ROW and kept separate from excavated material and shall not be used for padding or backfill the trench. Upon completion of backfill and ROW restoration, the topsoil material shall be returned in the area from where it was removed.

9.2.5. Grading of Right of Way

- After the ROW has been cleared, CONTRACTOR shall grade it, to enable pipeline construction. CONTRACTOR shall grade the ROW to remove sharp high points to minimize bending. Where the ROW passes through or along roads, tracks, pole lines, pipeline easement or any other improved or confined areas, CONTRACTOR shall grade only the width of the ROW necessary for the construction of the pipeline.
- No temporary or permanent deposit of any kind of material resulting from grading shall be dumped in the vicinity of roads, tracks, ditches, drainage ditches or in any other location where it might hinder traffic or water flow.
- Where CONTRACTOR encounters heaps of earth, either natural or artificial, or other materials on the ROW, these said materials shall be removed until the natural ground level is reached to ensure that construction of the pipeline trench is in stable ground.
- The disposal of said material shall be within the ROW or in safe areas appointed by CONTRACTOR and approved by EMPLOYER.
- At watercourse crossings, CONTRACTOR shall construct the ROW to maintain the original water flow conditions. When normal fording is not possible, temporary works shall be made to ensure regular water drainage to avoid overflows and to keep the existing morphology unchanged. Such works shall be previously approved by the concerned Authorities Proprietors and/or EMPLOYER.
- During clearing and grading operations at water crossings and when crossing cultivated land, care must be taken that water flow will not be hindered. CONTRACTOR shall be held responsible for any claims resulting from this interference with surface and subterranean water flows.
- CONTRACTOR shall ensure the protection of the site against water, which might affect the proper execution of the work. He shall ensure the drainage of the water and shall construct drainage systems as required without causing environmental damages.
- At no instance shall the ROW grading operations involve cutting into protective embankment structures of any type without previous written authorisation from the governing Authority, Proprietors and EMPLOYER.
- For ROW. grading operations over any existing line or service within the ROW,

CONTRACTOR shall obtain from the governing authority or operating EMPLOYER and proprietors a Permit-to-Work on either side of such line or service. Additionally, CONTRACTOR shall provide Industry standard stress calculations to prove that any defined crossing point is suitably designed (concrete slabs / bridging) etc. and induced impact surface loads are such that no damage occurs to the buried facilities in question.

- CONTRACTOR shall ensure all drainage and run-off of rainwater, surface, or groundwater, on land crossed by the ROW. or an off-site land affected by the Construction Works is well maintained and damages caused by any grading operation shall be made good. Where water run-off leaves the ROW. sedimentation traps and/ or silt fencing shall be established to avoid damage to the land adjacent to the ROW. Such facilities shall be regularly maintained and cleaned of silt and sediment to assure their functionality.
- The ROW. shall be continuously maintained by CONTRACTOR until completion of the Works.

9.2.6. Areas outside Easement

- The lease for short-term temporary use of private or public land areas for the purposes of general work, pipeline make-up storage or temporary resident camps for labourers, shall be solely the responsibility of CONTRACTOR.
- For any areas outside the approved right of way easement CONTRACTOR shall be responsible to ensure that the necessary investigations are carried out with the results and recommendations complied with. These investigations include but are not limited to; health & safety environmental, archaeology, geo-technical, authorities and land EMPLOYERs. The necessary qualified specialists (e.g. engineers, geologists, etc.) shall be utilized to carry out such investigations. For archaeology, CONTRACTOR shall mobilize a qualified archaeologist with a minimum of 5 years' experience in his/her field to determine if there are any historical areas of interest or national importance along the proposed ROW.
- Lands to be utilized as gravel sources, quarries, borrow pit (for bedding/back-fill) and/or rock disposal sites for excavated material require the above investigations and clearances as well as clearance from Local government authorities having authority in the pertinent metropolis.

9.3. Excavation and Trenching

9.3.1. General

- In and/or adjacent to public roads, open trenches within 500 meters of populated areas shall not be left open overnight without prior approval from EMPLOYER. Additional measures may be required by local authorities and/or by EMPLOYER to safeguard the local populace from accidents, including but not limited, to 24-hour watchmen and floodlighting

- All open trench and bell-holes shall be kept free of water. CONTRACTOR shall supply all materials for this and will receive no compensation for materials not later recovered. The cost of de-watering of surface water shall be borne by CONTRACTOR.
- CONTRACTOR shall employ equipment and methods to keep the trench to the established depth and gradients so that pipes can be evenly bedded throughout their length and at the correct cover, regardless of the type of soil encountered and regardless of the depth of excavation necessary.
- CONTRACTOR shall furnish all equipment, materials and supplies necessary for the completion and maintenance of the trench, including shoring and sheet piling, dewatering and any other incidentals, for the completion of the work in an approved manner.
- All efforts shall be made to reduce to a minimum the required number of bends to lay the pipe to conform to the contour of the ground and maintain the minimum depth of cover. This can be accomplished by cutting the trench slightly deeper at the crest of ridges and by gradually deepening the trench in approaches to crossings of roads, trenches, rivers, watercourses, underground cables and pipelines.
- Wherever practicable, unnecessary bends shall be eliminated by operating the trenching machine or excavators at various depths to be determined by CONTRACTOR's Bending Engineer and not by grading the R.O.W. It shall not be acceptable to install any elastic bending where a properly formed cold bend would address the deflection angle required.
- Where required by the Landowners, Proprietors or Companies operating in the field, CONTRACTOR shall provide safe detours, bridges across the trench or backfilled section of trench to ensure access is always maintained.
- In transition areas between different types of soils, the trench shall be dug to attain the depth specified for the soils requiring the deepest pipeline cover.
- The finished trench shall be free of roots, stones rocks or other object which may damage the pipe or pipe coating.

9.3.2. Trench Dimensions and Profile

- The trench shall be excavated to provide a minimum depth of specified cover allowing extra depth for any padding required beneath the pipe. Access ramps to be provided at the start and end of each continuous section of open trench.
- At point where work on installed pipeline is to be performed later (i.e. tie-ins) sufficient extra excavation shall be made so that welding and other work can be made satisfactorily, around the pipe in the trench and access ramps/ steps provided.
- The bell-holes shall be kept free of water and secured against collapse, so that works in the

trench may be carried out under safe conditions for the personnel.

- The depth of cover shall conform to the following, excluding any crown above the ROW level.

9.3.3. Dry Land Areas

- The pipelines in dry areas, unless otherwise specified, shall have a minimum cover from the top of pipe to the graded surface ground level or to the natural ground level if it is lower on both sides of the trench of 1.0 m minimum.

9.3.4. Swamp Areas

- In swamp areas, the cover from top of pipe to the ground level or consolidated mud shall be 1.5m minimum, unless otherwise specified on construction drawings.

9.3.5. Special Locations

- Extra depth cover shall be provided at the following areas
- At pipeline and cable crossings
- At all cased or uncased road and track crossings
- Across areas with unstable soil
- In areas prone to erosion, such as watercourses banks, gullies, ponds;
- Where required by governing Authorities or Proprietors.

9.3.6. Excavation in Rock

- The term 'Rock' shall be deemed to include any hard compact material (other than concreted or paved surfaces of roadways, etc.) which cannot be removed by mechanical excavation, rippers or manual means and therefore requires the use of pneumatic drills or rock breakers, rock cutting machines or blasting.
- Where the bottom of the pipe trench is in rock, or has hard protrusions capable of damaging the pipe coating, the bottom of the trench shall be excavated for a further 200 mm across the full width, so as to accommodate the 200 mm minimum layer of padding material. Bedding material grain size shall not exceed 10mm as validated with appropriate screening equipment.

9.3.7. Stringing

- Pipe stringing shall be allowed only after the R.O.W. has been cleared and graded as specified on construction drawings. The CONTRACTOR shall offload and string pipes along the route of the pipeline using suitable working plan approved by EMPLOYER. Pipes shall be strung out to cause the least practicable interference with the use of the land, and at intervals, gaps shall be left between pipes as directed to permit the passage of stock and equipment across the right-of-way.

- CONTRACTOR shall string the pipes so that the remaining overlap at tie ins at the time of the tie in is at least 2.0m long to produce a useable off-cut which CONTRACTOR will re-bevel and move forward to the Pipe Stringing for re-inclusion in the pipeline in a timely manner without causing damage to the "pup" or it's coating.
- Coated and wrapped pipes and fittings shall not be placed on the ground but placed on well-padded timbers provided by the CONTRACTOR. The pipes shall always be adequately supported by the full effective width of the padded timbers, so as to ensure no indentation of the pipe coating and wrapping material.

9.4. Bending of Pipes

9.4.1. Field Bends

- Cold bends shall be made by the cold stretch smooth bending method using bending machines equipped with internal padded mandrel, padded bending shoes and drive rollers. Prior to construction, the equipment shall be tested and qualified, witnessed by EMPLOYER representative, for cold bending to verify that the pipes shall be bent in accordance with applicable codes and this specification. The bends shall have uniform bearing under the pipe and will maintain the required depth of cover and shall be neither in tension nor in compression when placed in trench.
- The wall thickness of the bends due to stretching shall not be less than the nominal wall thickness minus the manufacturing tolerance. Reverse, mitred or wrinkled bends shall not be permitted. Any pipe bends showing evidence of buckling or wrinkling shall be cut out and replaced. No combination of bends shall be allowed in a single joint of pipe.
- Bending shall be accomplished without damaging or disbanding the pipe coating. Damage to the pipe coating due to bending shall be repaired in accordance with the requirements of Line Pipe Coating Specification. All completed field bends shall show no evidence of buckling, cracks, or wrinkling.

9.4.2. Elastic Bends

- Elastic bends shall have a minimum radius of 500 times the pipeline diameter (DEP 31.40.00.10-Gen). Bends of smaller radii shall be performed by cold bending.

9.4.3. Coating Repairs

- Defective areas of corrosion coating and concrete weight coating shall be repaired in accordance with Line Pipe Coating Specification. In any event repaired areas shall be subject to inspection and testing in accordance with the above specifications.

9.5. Line-Up and Welding

9.5.1. Checking and Cleanliness of Pipes and Fittings

- Every effort shall be made by CONTRACTOR when the pipeline is under construction to keep it free of dirt, waste, and other foreign matters and when work is not actually in progress all open ends of the pipeline shall be closed with an approved cap or plug, securely fixed to resist unauthorised removal.
- The inside of each pipe and fitting shall be individually checked by CONTRACTOR for cleanliness prior to lining up and jointing. All loose mill scale, shot or grit left in the pipes as a result of blast cleaning work and all other extraneous matter shall be removed by swabbing out with a pull-through, and then blown through with compressed air, using a long lance with radial jet type cleaning nozzle. Any pipe numbering labels shall be removed from inside the pipes, only after CONTRACTOR has recorded all necessary pipe information and transferred same to outside of pipe using permanent markers and characters at least 50 mm high. Pipe end caps shall be replaced and retained in position as long as practicable before joining. All dents containing creases, sharp edged imperfections of gouges shall be considered as defects.

9.6. Welding

CONTRACTOR shall align and weld together the pipe joints into a continuous pipeline. All welds in the pipeline made by CONTRACTOR shall be of the strength equal to that of the pipe as a minimum and shall in general be stronger than the base material itself.

Welding of pipelines shall be in accordance to the projects technical specification for welding of pipeline, DEP 61.40.20.30-Gen and ISO 13847:2000.

Prior to the performance of any welding the Construction CONTRACTOR shall submit the detailed welding procedure specification(s) (WPS) including sketches and diagrams for qualification and approval by EMPLOYER. Welding personnel shall also be qualified for the procedure. As a minimum a WPS shall contain the following information:

- Specific welding process or combination of processes;
- Specific welding technique or combination of techniques including direction of welding;
- Material specifications of the components, which will be welded;
- Details of weld joint configuration giving the shape of the groove, angle(s) of bevel, root face dimensions and allowable tolerances;

- Dimensions of test weld pieces and range of pipe sizes covered by same;
- Position of test piece (e.g. 6GR), welding positions covered, e.g. horizontal, vertical;
- Details of welding equipment, stating brand, type, voltage, current etc;
- Details of filler metal, giving trade name, classification and size of electrode or filler wire and chemical composition for each weld pass; Details of shielding gas (including backing gas), type, composition, flow rate and pressure;
- Maximum allowable misalignment between abutting surfaces;
- Method of cleaning procedure prior to welding;
- Details of procedure for control and documentation of welding consumables;
- Electrical characteristics of welding, i.e. current and voltage range, polarity for each size of electrode or filler wire (If pulse-arc welding is proposed, the pulse ranges)
- Heat input range;
- Welding travel speed or electrode run-out length for each weld pass. Number and sequence of weld passes indicating whether they are single bead or weaved passes;
- Details of procedure for pickling and how to passivate completed welds (external surface only);
- Details of procedure and equipment for cutting and re-bevelling;
- Details of stress relieving (if any)

During welding, the opposite end of the pipe joint is closed to reduce the internal blow. At the end of each working day, CONTRACTOR shall remove all waste generated during the day to an approved waste disposal area and install approved end caps on all open ends of welded sections and these shall not be reopened until work is resumed.

Upon removal of the cap, the line shall be visually inspected and, if necessary, swabbed to remove all dirt or foreign matter from the inside of the pipeline prior to welding operations. CONTRACTOR shall be responsible for any obstruction inside the pipeline.

End caps shall be placed using suitable techniques approved by EMPLOYER, to prevent damage to the bevels at the ends of the pipeline or welded section of line. Tack welding of end caps shall only be considered as a final measure if end caps are continually stolen. To permit clarity on when end caps shall be used, a pipe section requiring end caps shall be considered as being two pipes or more within any given welded pipe string.

9.7. NDT Inspection

All field welds shall be inspected by radiography. The NDT inspection shall be carried out by an independent Sub- CONTRACTOR who shall not be affiliated to CONTRACTOR. The NDT Sub-CONTRACTOR and his personnel shall be approved by EMPLOYER.

The approved Sub-CONTRACTOR shall provide the necessary equipment and qualified personnel to perform the work. CONTRACTOR shall be directly responsible for the approved Sub-CONTRACTOR.

CONTRACTOR shall fully co-operate with EMPLOYER and provide the following information and services, or any other assistance that EMPLOYER may reasonably require in order to:

- Keep inspection personnel informed of his daily schedule for welding and subsequent operations, particularly with regard to tie-ins and welds not in the main active work areas. This will be in the form of a daily work sheet;
- Make such welds fully accessible by maintaining access roads and the ROW in a proper suitable condition for use by EMPLOYER 's vehicles and, where necessary, providing suitable transport and/or towing facilities for EMPLOYER's personnel and vehicles;
- Provide space and enclosures for inspection equipment, supplies and, where specified, for the development, processing and viewing of radiographic films.

Extent and type of testing required shall be as stated in the relevant engineering specifications, drawings and other contract documents.

9.8. Coating of Field Joints

Field joints coating of PE coated pipes shall be made with heat shrinkable sleeves. Materials and application shall be in accordance with projects technical specification for heat shrinkable sleeve and DEP 30.48.00.31-Gen - Protective Coatings for Onshore and Offshore Facilities and the Line pipe coating specification.

9.9. Inspection of Coated Pipes and Fittings

High-tension electric holiday detectors with ring (rolling spring type) and brush electrodes shall be passed over all coated surfaces of pipes and fittings in the line. For straight pipes, holiday detectors shall normally be used in conjunction with rolling-spring type electrodes. A brush-type electrode shall be available at all times and used when the use of the rolling-spring electrode is not practicable.

Immediately prior to lowering-in, the pipe coating, including main pipe coating, shall be inspected with a holiday detector. Any coating defects found shall be repaired in accordance with line pipe coating specification.

9.10. Painting of Above Ground Pipeline Section

Pipeline, steel structures and supports, installed above ground shall be painted in accordance with project specification for painting.

9.11. Lowering-in the Pipe

Before starting the work, CONTRACTOR shall submit a lowering-in procedure for the approval of EMPLOYER. Prior to lower in of any pipe section CONTRACTOR will ensure that trench is free of water (Unless in an agreed swamp area), trench bottom is clean and free of any materials that could be harmful to the pipe or it's protective coating.

CONTRACTOR shall provide a sufficient number of side-booms, cradles, slings and equipment to support the pipe during lowering in.

The pipeline sections shall be lowered without interruption, avoiding contact with trench walls and impact on the bottom of the trench. Should it be necessary to interrupt the lowering-in operations, all equipment shall remain in place thereby continuing to support the pipeline in its correct position. The lowering crew shall excavate around the slings, if necessary, to clear them and to prevent dragging against the coating.

Horizontal bends shall be lowered so that the bend section is at a distance of at least 0.3 m from the trench walls.

When the lowering is achieved, the lifting bands shall be unfastened and recovered with particular care to ensure that the pipe coating is not damaged. CONTRACTOR shall use roller belts or other suitable belts which minimize the need for personnel to enter the trench to remove trench bed material to facilitate release of the slings

9.12. TIE-INS

9.12.1. General

- CONTRACTOR shall perform all work necessary to tie-in separate welded pipeline sections into a continuous system.
- CONTRACTOR shall bevel, align and weld the pipe in accordance with the requirements of the project particular welding specification.
- Welding after the pipe has been lowered into the trench (bell hole welding) shall be carried out only where sufficient clearance has been provided to enable the welders to exercise normal welding skill and ability in safety.
- In swamp and marsh areas the tie-in shall be carried out after lifting of the two pipe string ends out of the water, if necessary.
- CONTRACTOR shall carry out tie ins taking into account the minimum bending radius in accordance with project Stress Analysis Requirements.
- CONTRACTOR shall perform the tie-in as soon as practicable after lowering-in of the pipe strings and shall endeavour wherever practicable to undertake the tie-in at the hottest time

of the day. The overlap of the strings at tie-ins shall be a minimum of 2.0 m. When there is insufficient overlap pup pieces with a minimum length of 1.5m shall be used for the tie-in.

- The use of more than one pup at the same tie-in location shall not be permitted.
- All cutting shall be performed by an approved pipe cutter or thermal cutting and bevelling machine. Manual cutting shall not be permitted.
- All pups 1.5m or greater in length that are cut off at the tie-in points shall not be allowed to be accumulated but shall be moved ahead daily, by CONTRACTOR, as the Work progresses and welded into the line. In no case shall more than three welds be included in an 8.0 m long pipe section.
- All tie-in welds shall be 100 percent NDT in accordance with the requirements of Section 7.7 of this specification. All tie-in welds shall be coated with heat shrinkable sleeves in accordance with project field joint coating specification.

9.12.2. Golden Welds

- Tie-ins of adjacent pipeline sections already pressure tested shall be carried out with a single golden weld whenever possible; should the pipe sections not overlap or have appropriate length single pups with a minimum length of 1.5 m, previously hydrostatically tested, shall be used to make the tie-ins. Golden welds shall be minimized. Any golden weld shall be subject to the 100% radiography, 100% UT and 100% MPI.

9.12.3. Tie-In of Special Items and Fittings

- Connections of pipes, special items, fittings and equipment with different thickness shall be bevelled in accordance with the requirement of applicable codes and specifications.
- Only machined transition pieces shall be used for joining pipes of different wall thicknesses. In case of constant outer diameter, recommended internal diameter change is progressive with a maximum ¼ slope transition. Tapering of the ends of special pieces will only be carried out upon EMPLOYER authorisation.

9.13. BACKFILLING AND REINSTATEMENT

9.13.1. Preliminary Backfill

- Excavated material (other than topsoil or surface materials) shall be used as soft surround for the installed pipe. This material shall be specially selected and sieved if necessary, so as to be free of stones, pieces of rock, timber, stocks, roots, debris and any other material which may damage the pipe coating and/or initiate bacterial corrosion.
- If the excavated material cannot be rid of objectionable materials such as indicated above, thereby making it unsuitable for placement around the pipe, CONTRACTOR shall locate

and transport suitable material (which meets the requirements of this specification) in sufficient quantities to completely surround the pipe and cover it to a depth of at least 200 mm. Non suitable excavated materials for backfilling must be evacuated immediately and not stored along the pipeline ROW.

9.13.2. Backfilling

- No rubbish, vegetable growth or other perishable materials shall be put into the pipe trench along with the backfill material.
- The remaining excavated material shall be returned to the trench in layers not more than 200mm and thoroughly compacted by mechanical plate vibrators or rammers.
- Surplus excavated sub-soil shall be removed from the site, or spread evenly over the working width. When spread over the working width, the topsoil which shall have been removed and set aside shall be spread on top of the sub-soil backfill.
- Any excavated material which, if used, might cause damage to the pipeline coating shall be considered as unsuitable for backfill purposes. This material shall be removed from site immediately by CONTRACTOR. The maximum stone size to be returned to the trench shall be 200mm.
- In carrying out backfilling operations, CONTRACTOR shall make due allowance for subsequent settlement and shall promptly make good any settlement that takes place up to the end of the maintenance period.
- The backfill material shall be crowned (heaped up) along the trench line to a height of between 200 and 300 mm above the adjacent ground surface.
- Where tracked equipment is used for backfilling operations, such equipment shall not be allowed to travel along, across or over the installed pipe unless such loads can be proven by calculation to not detrimentally affect the pipeline integrity.
- Where the use of borrowed fill is necessary to crown the backfill to the required height, the crown fill shall be compatible with the earth existing at the location, free of weeds and rubbish and shall be placed in a manner that would not obstruct the free flow of water. All terraced levees and sides of drainage or irrigation canals that are cut by the pipe trench shall be backfilled in 150 mm thick layers and each layer thoroughly compacted by hand or machine, so as to provide a good bond between the undisturbed sides of the ditch and the new backfill material.
- On sloping ground, diversionary furrows or levees shall be provided across the pipe trench backfill, to direct the flow of rainwater and flooding into the natural drainage courses and away from the pipe trench. However, the surface drainage water must not be diverted into channels other than those it followed before the pipeline was laid, except in cases where its

diversion offers protection to the pipeline.

- Particular care shall be exercised to ensure that all drainage ditches are properly reinstated and left unobstructed, so as to prevent the backing up of water and flooding. Where water run-off could occur from the R.O.W. sediment traps and silt fences shall be installed and adequately maintained throughout their life on the project.

9.13.3. Support against Settlement

- Where the pipeline emerges above ground as shown on project drawings, and at other locations where the ground/soil conditions indicate possible settlement, special care shall be taken to ensure that the buried section of pipeline is adequately supported against settlement. The pipeline shall be protected with rock shielding supported on bags of lean mix concrete or by such other method called for in construction drawings.

9.13.4. General Reinstatement

- CONTRACTOR shall reinstate all areas disturbed (either directly or indirectly) by the construction of the pipeline, to a condition at least equal to that existing before the commencement of the works and shall provide all materials required to complete these reinstatement works. Reinstatement shall apply to all access roads, kerbs, footpaths, hedges, walls, fences, ditches and land drains etc., that are affected by the Work. All such reinstatement shall be as per the requirements of this specification.
- Reinstatement shall be carried out as soon as practicable after completion of backfilling. All flume pipes, shoring, sheet-piles shall be removed from the site at the latest during the reinstatement phase.
- Generally, in backfilling the pipe trench, all the surface material that was stripped and set aside prior to excavation shall be replaced in its original position as far as possible, together with such new materials as may be required in order to reinstate the land to a condition at least equal to its original state before CONTRACTOR entered onto the land.
- CONTRACTOR shall remove all surplus excavated material (other than topsoil), scrap, and rubbish from the working width and leave the whole R.O.W. in a tidy condition upon completion of the works. CONTRACTOR shall ensure that no harmful materials are left. No rock pieces, caused during trenching / blasting operations, larger than 60mm shall be permitted to remain within the R.O.W. after reinstatement.
- CONTRACTOR shall be responsible for making good all settlement, scouring, erosion or other defects until the expiry of the maintenance period.
- Punch lists shall be developed by EMPLOYER for CONTRACTOR action in respect to any outstanding reinstatement works after CONTRACTOR confirms his works are completed. All

punch list items shall be closed out by CONTRACTOR in a timely fashion to avoid impact to handover date.

9.13.5. *Cultivated Land Reinstatement*

- After completion of the backfilling, the previously separated topsoil shall be replaced over the stripped areas to its original full depth. CONTRACTOR shall provide all necessary additional topsoil that may be required to replace any of the original material that has been lost or dispersed.
- Topsoil shall be left in a loose usable condition ready to receive seed and shall be free of stones as the adjacent land.
- CONTRACTOR shall also carry out all harrowing, raking and watering of previously cultivated land necessary to achieve its proper reinstatement over the full working width, and leave it ready for replanting and break up compaction in the transit areas.
- CONTRACTOR shall maintain the trench reinstatement and make good all settlement, washouts, and defects until the expiry of the maintenance period.

9.13.6. Storage Areas, Working Areas and Accesses Reinstatement

- CONTRACTOR shall reinstate the surface of all storage areas, working areas, access routes and footpaths to a condition at least equal to that existing before the commencement of the Construction Works.

9.14. Pipeline Markers and Warning Signs

- Pipeline markers shall be installed every one kilometre (1 km) and shall be suitable for easy observation by ROW patrols. Pipeline markers shall be placed in such a manner that a marker is located at every pup joint, pup joints shall be installed at 2 km intervals. In addition, pipeline markers shall be installed at the following locations:
- On both sides of crossings with creeks, rivers, public roads, railways, above ground appurtenances, third party services, field boundaries, etc.;
- At any significant horizontal deviation of the pipe routing;
- At any other location as deemed necessary or required by the EMPLOYER or authorities having jurisdiction.
- At all Cathodic Protection test points/stations.
- Coordinates of the pipeline markers shall be taken and documented. Pipeline markers and their respective coordinates shall be shown on the as-built alignment sheets.
- Each marker must be visible from the previous and next marker location and be tall enough to be visible above the local vegetation. Such markers shall be suitable for easy observation by ROW patrols. The markers shall carry as minimum, basic information, such as the name

of EMPLOYER, pipeline service high pressure, etc. as per typical markers and warning signs drawings. Emergency Phone Numbers, high pressure gas pipeline and other key information shall be installed on the marker posts.

9.15. Hydrostatic Testing and Pre-commissioning

- Hydrostatic testing and pre-commissioning shall be done according to the requirements of the projects Pipeline Cleaning, Gauging, & Hydrotesting SpecificationEA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-010
- The Construction CONTRACTOR shall make accessible to the EMPLOYER all portions of the pipe laying operation so that each phase may be thoroughly inspected in accordance with the contract.
- Four inspections are required as a minimum:
    - Inspect the condition of the trench bottom just before the pipe is lowered in;
    - Inspect the surface of the coated pipe as it lowered into the trench;
    - Inspect the fit of the pipe to the trench before backfilling; and,
    - Inspect all repairs, replacements or changes ordered before backfilling

9.16. ADDITIONAL REQUIREMENT FOR CONSTRUCTION IN SWAMP AREAS

9.16.1. General

- This section is intended to be read in conjunction with all the foregoing sections of this specification, but where conflict with the particular requirements of construction on land occurs, the specific requirements of this section shall prevail.
- CONTRACTOR shall take special care that traffic lanes in open water and in all rivers and creeks are kept open for traffic and shall warn such traffic of the movements of large floating pieces of construction equipment and the placement of equipment anchor lines.
- In swamp areas, pipelines shall be laid to the spacing and depth as specified herein unless otherwise specified on construction drawings.
- No bridging or spanning of creeks, canals or ditches shall be permitted.

9.16.2. Depth of Cover

- The minimum depth of cover above the top of the pipe for pipelines buried in swamp areas shall be 1.50 m minimum from true ground level or the lower of the original ground level or graded surface as per the projects trench configuration details.

9.16.3. Excavation of Pipeline Trench

- The pipe trench shall follow the route surveyed and staked out beforehand.

- CONTRACTOR shall be responsible for the setting-up of all necessary line, grade, and range poles. Such poles shall be of a substantial nature, conspicuously painted and firmly set in the ground. CONTRACTOR shall be responsible for regular soundings and surveys of the trenching operation, the results of which shall be submitted to EMPLOYER for information. The pipeline trench shall have the minimum dimensions as indicated in the trench configuration details.
- CONTRACTOR is requested to propose the less aggressive environment method to perform the excavation works.
- Where the pipe is to be installed by the 'push-pull' floatation method, the pipe trench shall be wide enough and deep enough to accommodate buoyancy tanks and/or other necessary flotation equipment.

9.16.4. Lowering or Sinking Pipe Into Trench

- Whichever method is used for lowering or sinking the pipe into the trench, it shall be performed in such a manner that the coating is not over stressed due to excessive deflection or in any other way damaged. The pipe shall lie naturally and be continuously supported along its entire length in the bottom of the trench.
- Where the flotation method is used, the pipe shall be floated into position fitted with adequate floatation tanks or pontoons, as necessary. Care shall be taken to guide the pipe through the trench and prevent the pipe from rubbing against the wall of the trench.
- CONTRACTOR shall use a sufficient number of floatation tanks or pontoons and take all other precautions necessary to eliminate stresses in the suspended pipe and not exceed the specified minimum bending radius limitations.
- After the pipe is in position, the floatation tanks shall be removed in such a manner as not to damage the coating or endanger CONTRACTOR personnel releasing them.
- If CONTRACTOR elects to use the push/pull method, he shall submit to EMPLOYER fully detailed procedures including a detailed list of equipment and maximum pull force he proposes to use.

9.16.5. Backfilling

- Backfilling of the trench shall be performed with excavated material resulting from the trenching as per requirements defined in Section 11
- Backfilling material shall be approved by EMPLOYER. Backfilling material shall be selected on the basis of the following criteria:
- The pipeline settlement under the weight of the backfill material shall be acceptable,
- The upheaval buckling shall be avoided.

- Natural backfilling for areas subject to potential upheaval buckling is prohibited.
- Where tracked equipment is used for backfilling operations, such equipment shall not be allowed to traverse along, across or over the installed pipe. Where specified crossing locations are needed by CONTRACTOR across third party pipelines adequate crossing methods shall be proposed for EMPLOYER approval along with stress calculations to assure integrity of the buried line.
- In each case where CONTRACTOR has no option but to cross already installed lines, he may do so only when adequate precautions have been taken to protect the pipeline.
- If CONTRACTOR needs to open channels across swampy areas for passage and flotation of construction equipment such as dredging or pipe laying barge, CONTRACTOR shall perform the backfilling of the pipeline trench only. The channels shall be kept open.

10. Quality Assurance

The CONTRACTOR shall show that an effective system of Quality Assurance is in operation, for both products and services.

The CONTRACTOR shall submit, with his tender documents, a copy of his Quality Manual for review by EMPLOYER. The Quality Manual should contain but not be limited to the following:

- A signed policy statement on commitment to quality by the head of the CONTRACTOR;
- An organisation matrix of the CONTRACTOR indicating reporting responsibilities;
- An index of the CONTRACTOR quality procedures;
- A brief outline of each procedure indicating individual responsibilities for maintaining quality.

The CONTRACTOR shall also submit a typical Quality Assurance Plan (QAP) and Quality Control Plan (QCP) with his Tender documents. Quality documentation required after award of contract is detailed in the Purchase Order.

The CONTRACTOR shall ensure that a full time QA/QC person is available to monitor the following:

- Material certification
- Traceability of materials
- Welding procedures
- Welder qualification
- Field welding
- NDT Procedures
- Trench depth
- Sand padding
- Backfilling, compacting & reinstatement
- As-built documentation
- Non-conformance reports
- Document control
- Project QA/QC Dossier

11. Crossings

11.1. General

Principally crossings shall be constructed by the open cut trench method. Alternative methods i.e., horizontal directional drilling may be utilized subject to prior approval from the EMPLOYER and authorities having jurisdiction and Landowners.

Before start of construction activity, the survey crew shall mark with visible signals the position of any pipelines in the area of 150 m left and right of pipelines route alignment.

Separate as-built drawings and reports shall be prepared for each individual crossing for submission to the EMPLOYER, authorities having jurisdiction in the area, Landowners and EMPLOYERs / operators of foreign services as appropriate.

If strings of pipe are pre-fabricated for crossings the strings shall be pressure tested and only accepted pipe strings shall be placed at their final location. Pressure test shall be repeated after placement of the pipe strings at their location. Crossings shall only be backfilled after successful completion of the pressure test. For pulling of strings winches of sufficient capacities shall be utilised for the Construction.

11.2. Major Road Crossings

11.2.1. General

- Pipe crossings beneath roads shall be installed with bored or open cut methods in accordance with the requirements of authorities having jurisdiction over the crossing and applicable construction drawings.
- Road crossing shall be performed in accordance with the pipeline road crossing drawing.
- Care shall be taken to minimise inconvenience to traffic while such crossings are being installed and CONTRACTOR shall furnish all necessary warning signs, safety barriers, watchmen and lookouts to warn traffic and safeguard self and EMPLOYER, his employees, other involved parties, and all members of the public by day and by night.
- Suitable measures shall be used at all times to prevent damage to roads when crossing with construction equipment. Rubber tyres or equivalent means shall be used.
- CONTRACTOR shall be responsible for making good any damage to the road to the satisfaction of the road Authority or Proprietors. All costs for repairs shall be to CONTRACTOR charge.
- As-Built drawings of all road crossings shall be submitted within 30 days of completion of the road.

- The shoulders, trench banks and slopes and pavement of all roads crossed shall be restored to their former condition. Such restoration shall be as required by the authority having jurisdiction.

11.2.2. Bored Case Crossings

- Prior to commencement of any boring, CONTRACTOR shall submit to EMPLOYER complete details of the proposed construction method for approval.
- To install the casing pipe with the boring machine, pits are required at both sides of the crossing.
- The dimensions of the pits shall be determined taking into account the space required for setting out the boring equipment and laying of the pipeline to the specified profile.
- The pit sides shall be suitably sloped to prevent cave-in or landslides. To this purpose it may be necessary, especially when the water table is present in the trench, to install sheet piling against the pit walls suitably sized to resist the earth thrust.
- Thrust boring shall be performed at a minimum depth of 1.50 m below the lowest part of the road formation. Before thrust boring commences assurance of alignment across the entire length of the bore shall be required by EMPLOYER, subject to sub-soil condition evaluation which will be defined whilst excavating the thrust and reception pits.
- Thrust boring shall commence at a distance not less than 10 m from the near edge of carriageway and ends at the same distance from the far edge of the carriageway.
- As-Built drawings of all road crossings shall be submitted within 30 days of completion of the road.

11.2.3. Open Cut Road Crossing

- When the trench is cut across roads, temporary crossings of adequate strength or diversions shall be provided to allow the passage of normal traffic with a minimum inconvenience and interruption. CONTRACTOR shall evaluate each situation and submit the proposed solution to EMPLOYER for information and to the Authority for approval. As-Built drawings of all road crossings shall be submitted to EMPLOYER within 30 days of completion of the road.
- The excavation of the trench across the road shall commence only where the section of pipeline is tested and ready for installation. CONTRACTOR shall place and maintain day and night warning signs and barricades throughout the period of the work and in accordance with the requirements of the authority having jurisdiction. If necessary, CONTRACTOR shall provide watchmen at his cost until the road is restored.
- If the road has a bitumen or asphalt surface, the anticipated edges of the excavation are to be cut with an asphalt saw prior to excavation to avoid tearing of the surface beyond the

excavation limits.

- The pipe trench shall be carefully excavated so that pipe can be evenly bedded throughout its length. Road metal or other surface materials and hardcore shall be kept separate from other excavated material.
- The finished trench shall be free of roots, stones, rocks or other hard objects which may damage the casing pipe or pipe coating. If necessary, provision shall be made for dewatering.
- The trench width on open cut crossings shall be at least 600 mm wider than the outside diameter of the pipe.
- Where possible, CONTRACTOR shall complete trenching, lowering-in and backfilling of the crossing and remove any temporary bridging before the end of the working day.
- Adequate length of pipe shall be left uncovered on both sides of the crossing to allow for the crossing to be tied into the mainline. Wherever possible the crossing shall be installed prior to mainline lower in.
- Backfilling of the crossing shall be completed with approved selected material having the same characteristics as the existing material. Backfilling shall be carried out in layers not exceeding 150mm and shall be carefully compacted by the use of mechanical tampers so that each layer has a density equal to or greater than the adjacent original material.
- In the circumstance that permanent reinstatement cannot immediately be commenced, CONTRACTOR shall carry out temporary reinstatement of the road surface within three days after backfilling. Temporary reinstatement work shall be carried out to a manner, which will permit its inclusion in the final reinstatement.
- To carry out the permanent reinstatement, CONTRACTOR shall remove the temporary surfacing by cutting back to a neat vertical edge in the original road surfacing on either side of the trench. Then sub-base shall also be removed to a depth necessary to accommodate the permanent reinstatement. The surplus materials arising from both operations shall be disposed of. The loosened surface of the sub-base shall then be thoroughly recompacted, using mechanical tampers.
- Immediately before the placing of permanent reinstatement material, the edges of the original road surfacing shall be coated with a suitable grade of hot bitumen.
- Permanent reinstatement material shall be compacted by mechanical means; degree of compaction of each layer of material shall be 95 percent minimum of modified Proctor, unless otherwise specified on construction drawings. CONTRACTOR shall ensure that the final reinstated surface is finished flush with the original road surfacing.
- The permanent reinstatement of roads shall be at least of equal standard as the original.

11.2.4. *Secondary Road and Track Crossings*

- Secondary roads and tracks shall be crossed by the open cut method and without casing pipe, with a 1.50m minimum cover top of pipe as shown in the pipeline road crossing drawing.

11.2.5. Crossing of Underground Services

- The presence and location of other buried pipelines, water mains, sewers, cables, and other underground structures which cross or are adjacent to the pipeline being constructed shall be established. Prior to commencement of any crossing, CONTRACTOR shall notify the Proprietor or Authority having jurisdiction in sufficient time (at least seven days ahead) to ensure the presence of their representative and where required by EMPLOYER, a Proprietor / Authority permit is in place.
- Underground installations shall be completely exposed by hand excavation, at least 0.5 kilometre ahead of any mechanical excavation work. Such excavation shall be sign posted and fenced in a suitable manner until the trench is dug. No mechanical excavation shall be allowed closer than 2m of any such underground installation or as specified by Proprietor or Authority having jurisdiction of the underground installation.
- Unless otherwise instructed by EMPLOYER, the pipe to be laid under such underground installations may necessitate additional trench excavation depth to provide the minimum free clearance between the pipeline and the line or utility being crossed.
- The minimum vertical separation between the pipeline and existing facilities shall be 0.60m in dry land and 1.20m in swamp areas.
- All existing pipelines or utilities crossed by the pipeline shall be adequately supported in accordance with the approved method.
- Prior to exposure of any existing pipeline a sign and substantial barrier shall be erected to prevent any damage to the pipelines or services. This is particularly important where there are heavy vehicles working in the vicinity and the possibility that removal of the bund for trenching operations opens up an unauthorised vehicle crossing.
- CONTRACTOR shall supply all equipment and materials necessary for the completion and maintenance of the trench, including shoring, sheet piling, de-watering and any other incidentals for the completion of Construction Works.

11.3. River Crossing

11.3.1. General

- Where the route of pipeline Crosses River, creek, channel or other waterway crossings the pipeline shall be buried in the natural bed of the watercourse and constructed in accordance with pipeline River crossing drawing prepared by CONTRACTOR and approved by EMPLOYER.

- No spanning or bridging of rivers, creeks, steams, ditches, or any other permanent or seasonal watercourses shall be permitted.
- Before commencement of work at a crossing, CONTRACTOR shall submit to EMPLOYER for information, proposals for preventing any erosion of the crossing banks during and/or after the construction stage.
- In carrying out water crossings, CONTRACTOR shall comply with all regulations imposed by the Nigerian Port Authority, Inland Waterways and other Authorities as appropriate.
- CONTRACTOR shall take special care that traffic lanes in open water and in all rivers and creeks are kept open for traffic without being interfered with by the movements of the floating pieces of construction equipment and the placement of their anchor lines.
- Any obstruction which might create a navigational hazard shall only be allowed by permission of Local, Federal or other Government Agencies and shall be marked and maintained in accordance with instructions issued by these Agencies. These instructions shall be passed to EMPLOYER for information.
- All dredging shall be carried out in accordance the requirements of the Nigeria Ports Authority and the Inland Waterways as appropriate.

11.3.2. Depth of Crossings and Cover

- At water crossings the trench across the bed of the watercourse shall be made deep enough to provide the minimum depths of cover over the crown of the pipeline given below, unless otherwise called for in the approved working drawings.
- For trunk pipelines and delivery lines, the installed depth of the crossing shall be as specified by the Water Authorities, or to the minimum depth (cover) whichever is the greater over the crown of the pipe as follows:
- Crossings of small watercourses of depth less than 0.5 m, depression liable to flooding and ditches which are not the subject of special attention shall be treated as normal pipe laying except for the cover that in no case shall be less than 1.2 m below the lowest cleaned bottom bed level over the full width of the crossing or as indicated on construction drawings.
- At crossings of creeks, canals, ponds and minor rivers of width less than 60 m and depth greater than 0.5 m, the pipeline shall be laid to a minimum depth of 1.5 m from lowest bed level below scour depth, over the full width of the crossing or as indicated on construction drawings.
- At crossings of rivers or creeks of width larger than 60 metres, the minimum cover of pipeline

shall be 4.5 m below LLWS or 1.5 m below the lowest bed level below scour depth whichever results in the greater depth or as indicated on construction drawings.

11.3.3. Surveying of the Crossing

- CONTRACTOR shall take depth soundings of all creek and river crossings. The average mud level in the vicinity shall be determined and established by a pegged bench mark.
- CONTRACTOR shall subsequently set a tide gauge at the edge of the ROW with a zero reading at the established mud level. The depth of the creek or crossing from the mud level shall be recorded for the deepest part of the creeks and rivers within 3 metres on each side of the centre line of the pipe trench.
- The pipeline profile of the crossing shall then be verified by CONTRACTOR taking into account the following parameters:
- Minimum allowable bending radius specified or shown on the approved construction drawings (the radius of the sag/over bends at the banks of the watercourse crossing shall be in accordance with the applicable requirements of section 13.0 of this specification, unless otherwise specified or shown on the approved construction drawings).
- Possible settlements or snaking of the pipe after laying.
- Required depth of crown of pipe below LLWS.
- Required depth of cover after backfill.
- Unless otherwise stated in the CONTRACTOR construction procedures, specifications or specific approved construction drawings, all pipeline crossings shall be laid stress free and the pipe shall be continuously supported along its entire length and lie naturally on the bottom of the trench in both the approaches to the crossing and in the bed of the water crossing.
- As-Built Drawings for all watercourse crossings shall be submitted to EMPLOYER within 30 days of crossing installation.

12. Installation of Appurtenances

12.1. Valve and Pigging Stations

Valve and scraper trap assemblies shall be installed at proper locations as defined on the general arrangement drawings for above ground facilities. During welding operations valves shall be in the fully open position. Pig traps shall be installed on proper steel support at elevation and orientation specified on construction drawings.

13. Cathodic Protection System

13.1. Installation of Isolating Joints

Isolating joints shall be installed at the locations shown on the construction drawings. Handling and installation of the isolating joints shall be carried out with all precautions required to avoid damage and excessive stresses. The original pup length of the insulating joints shall not be reduced.

CONTRACTOR shall attach jumper cables to the joints prior to installation to avoid damage by voltage passing through them in the event of incorrect placement of the earth probe during welding.

During the execution of in-line connection welding, the propagation of heat shall be avoided. The isolating joints shall be tested and inspected by CONTRACTOR before welding into the pipeline and then repeated after welding of the joint into pipeline to verify that the assembly is undamaged and in proper service. All tests shall be conducted by a person competent in cathodic protection commissioning, documented and signed off by EMPLOYER. All test results shall be transmitted to EMPLOYER for record.

13.2. Test Point Installation

CONTRACTOR shall submit a cathodic protection method statement to cover all cathodic protection installations for EMPLOYER approval. CONTRACTOR shall employ a suitably trained cathodic protection technician to perform all installation work.

Cathodic protection (CP) of pipeline shall be impressed current system in accordance with the project cathodic protection calculation, specification and the relevant drawings.

CONTRACTOR shall furnish all material and install the complete cathodic protection system. Cathodic protection test posts shall be installed in accordance with Cathodic Protection Installation Details drawings. Cathodic protection cables shall be Pin Brazed to the pipelines and properly identified before backfilling for later connection to the test posts. After the cable connection is completed the coating shall be repaired with an approved repair material.

After backfilling, CONTRACTOR shall check all test points according to the approved procedure, for continuity, and those found broken or otherwise defective shall be repaired.

CONTRACTOR shall install test stations as defined on the pipeline alignment sheets and cathodic protection test facility schedules generally in accordance with the following:

- Every kilometre along the pipeline

- At temporary cathodic protection installations
- At principal road crossings
- At river crossings
- At foreign pipeline crossings
- Each side of steel cased road crossings
- At monolithic isolation joints
- At block valve stations

CONTRACTOR shall maintain and update on a weekly basis the test facility schedule for each pipeline and submit these to EMPLOYER for review.

At locations where the pipeline crosses or ties-in to existing buried metallic foreign services CONTRACTOR shall liaise with the foreign service operator and secure in writing the operator's specific requirements for cable connections / bonding and the repair to the coating on the foreign pipeline.

13.3. Temporary Cathodic Protection System

During the installation, all pipes shall be protected against corrosion with a temporary cathodic protection (CP) system.

The Temporary CP system shall provide effective protection at least until such time as the permanent CP is fully operational. The temporary CP shall start when the permanent CP system cannot be achieved within 30 days of pipe burial for all individual sections prior to tie in. CONTRACTOR shall propose to EMPLOYER within his procedure for EMPLOYER approval all temporary CP materials and equipment to be used to protect the pipeline.

13.4. DCVG Inspection

A DCVG survey shall be performed on all buried pipe sections within 7 days of any pipe being installed and backfilled. CONTRACTOR shall produce a detailed procedure for

EMPLOYER approval defining all the DCVG works to be undertaken. A qualification to determine acceptable defect size/s shall be proposed by CONTRACTOR and approved by EMPLOYER and such qualification shall be undertaken on a pipe or part thereof in an in-situ environment. Allowable defect size shall be approved by and at the discretion of EMPLOYER.

14. Documentation

CONTRACTOR shall perform the survey (i.e. topographical survey) to collect the as-laid data of pipeline and appurtenances. A Survey of all welded pipe joints to be made to determine final cover meets specified depth of cover requirements.

Information regarding sections of pipe out of compliance in relation to alignment or cover must be passed to EMPLOYER immediately and no backfill to be carried out on the affected section until the problem is resolved to the satisfaction of EMPLOYER.

Prior to the start of the as built activity CONTRACTOR shall prepare and submit, for EMPLOYER review, the procedure for the execution of field and office as built work. The procedure shall detail, as a minimum, the method of data gathering in the field, data processing and computation, as built documents preparation and issue.

14.1. Welding Log

CONTRACTOR shall gather all data and information for the preparation of the pipe and weld book. The format of the pipe book and the recording procedures shall be prepared by CONTRACTOR and approved by EMPLOYER.

CONTRACTOR shall produce for EMPLOYER approval a Weld History Tracking System. In said system shall also be all data applicable to each weld including but not limited to NDT & Field Joint Coating activities, current status of each and all previous history. To carry out such work, CONTRACTOR shall provide experienced personnel throughout the duration of the Construction Works.

The pipe and weld book shall contain as a minimum, the following data:

- Data relevant to the project and section thereof
- Sequential number
- Date of stringing
- Length brought forward (for pipes and other materials);
- Remaining Length
- Alignment sheet number
- Kilometre position
- Diameter of pipeline
- Length of each pipe
- Length of pipe per page
- Wall thickness
- Location of wall thickness changes (if any)

- Pipe number
- Location and angle of field bends, factory bends and in-line fittings
- Cut and renumbered pipe ends
- Coating
- Date of welding
- Pipeline Laying Direction
- Weld numbers
- Welder numbers and stencils
- Welding Procedures (including type of, electrode, diameter)
- Weld treatment
- Radiography equipment used
- Inspection Reports
- Limits of block valve stations, scraper trap stations, water crossings, road crossings, pipeline crossings, buried service crossings
- Test pressures, data and test (hydrostatic).
- Grade of pipe
- Heat Number
- Original pipe length if cut
- Tie in weld and golden weld position

In order to achieve this, all pipe elements shall be recorded. All pipe pups carried forward and all condemned pipe shall be tracked and documented for material accountability.

The book will include computer PC based Recordable Compact Disks and hard copy procedure records and reports for submittal to EMPLOYER via transmittal after completion of the work and prior to demobilisation.

14.2. Material Identification

For each pipeline element other than pipe, e.g. bends, valves, and other fittings, it shall be necessary that other than recording the factory provided serial number, heat number, pipe number etc, that other parameters of pipeline elements shall also be recorded in the above said list.

If a joint will be cut, before pipe cutting, CONTRACTOR shall paint serial number and stamp the pipe number on both side of the joints (which have been made by cutting) with a low stress die. The new data of the cut joints shall be recorded from CONTRACTOR and reported in the as-built documentation.

Pup pieces and joints with a diameter greater than half a metre shall be clearly marked up by painting on them the pipe number and cut piece progressive sequence.

The result shall be in such a manner that for all pup pieces and cut joints, of diameter greater than a half a metre, CONTRACTOR shall submit to EMPLOYER their traceability. Pipe which cannot be identified shall be removed and recorded before being stockpiled by CONTRACTOR.

Other As-built documentations to be submitted by CONTRACTOR are;

- Management summary;
- Construction report (narrative);
- Pre-installation survey report;
- Diary of events;
- As–built drawings (i.e. pipeline alignment sheets and manifold drawings);
- Daily, weekly and monthly reports;
- Deviation reports;
- Field Joint reports;
- Damage and repair records;
- Construction Equipment Calibration records;
- Safety reports (accidents / incidents);
- Data sheets, material certificates, etc. of permanent materials;
- Construction procedures, drawings, calculations;
- Gauging and hydrotest documents including test charts;
- Pre-commissioning / commissioning report;

15. Installation of Pipeline Crossing By Horizontal Drilling (Hdd)

This section is covered in another document "HDD Installation Specification EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-014"